

Data Dictionary

Hyperspectral Grass Reflectance Dataset

Dataset: Pau et al. (2025) - Grass hyperspectral reflectance and traits

Dryad DOI: 10.5061/dryad.gf1vhhn0n

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Purpose

This data dictionary defines all variables in the two main CSV files (spectra and traits). It's meant to help users understand what each column represents and how to interpret the values.

File 1: Pau_et_al_2025_spectra.csv

Structure: 426 rows × 2,151 columns

Row = one leaf sample

Column 1: Sample Identifier

Variable: sample_id

Type: Text/String

Description: Unique ID for each sample, links to the traits file

Format: Appears to be [SITE][SPECIES][NUMBER] (e.g., "KNZ_ANGE_001")

Missing values: None (required field)

Columns 2-2,151: Spectral Reflectance

Variable naming: X[wavelength]

Examples:

- X400.0 = reflectance at 400 nm
- X550.0 = reflectance at 550 nm
- X800.0 = reflectance at 800 nm
- X2400.0 = reflectance at 2400 nm

What the numbers mean:

- Reflectance values on a 0-1 scale

- 0 = complete absorption (no reflection)
- 1 = complete reflection (no absorption)
- Typical leaf values: 0.05-0.15 in visible, 0.40-0.60 in NIR

Spectral regions:

- **VIS (400-700 nm):** 600 bands - pigment absorption
- **NIR (700-1300 nm):** 1,200 bands - leaf structure
- **SWIR (1300-2400 nm):** 2,200 bands - water content

Missing values: About 1% missing, mostly in SWIR region around water absorption bands (~1400 nm, ~1900 nm). Coded as NA.

Important notes:

- Column names start with "X" because CSV column names can't start with numbers
- Wavelength spacing is 0.5 nm intervals
- SWIR region is noisier due to instrument sensitivity

File 2: Pau_et_al_2025_traits.csv

Structure: 426 rows × 23 columns

Row = one leaf sample (same samples as spectra file)

Linking Variable

Variable	Type	Description	Missing
sample_id	Text	Unique sample ID, matches spectra file	None

Taxonomic Information

Variable	Type	Description	# Categories	Missing
species	Text	Scientific name (Genus_species)	73 species	None
genus	Text	Taxonomic genus	48 genera	None

Variable	Type	Description	# Categories	Missing
subfamily	Text	Grass subfamily	5 subfamilies	None
tribe	Text	Taxonomic tribe	~15 tribes	~5%

Subfamilies: Chloridoideae, Panicoideae, Pooideae, Aristidoideae, Arundinoideae

Functional Classification

Variable	Type	Description	Categories	Missing
photosynthetic_pathway	Categorical	Type of photosynthesis	C3, C4	None
lineage	Categorical	Evolutionary grouping	Multiple	None
growth_form	Categorical	Growth habit	Various	~8%

C3 vs C4:

- C3 = standard photosynthesis (most temperate grasses)
- C4 = modified pathway, more efficient in hot/dry conditions

Site and Collection

Variable	Type	Description	Missing
site	Categorical	Site code (11 sites total)	None
collection_date	Date/Text	When sample was collected	~10%

Site codes: KNZ, CDR, SEV, SGS, HYS, plus 6 others (see README for details)

Morphological Traits

Variable	Units	Description	Typical Range	Missing
SLA	mm ² /mg	Specific Leaf Area - leaf area per mass	10-45	~15%

Variable	Units	Description	Typical Range	Missing
LDMC	mg/g	Leaf Dry Matter Content - tissue density	150-450	~12%
leaf_thickness	mm	Blade thickness	0.1-0.5	~18%
leaf_area	cm ²	Individual leaf area	5-50	~10%
leaf_length	cm	Blade length	10-50	~8%
leaf_width	mm	Blade width	2-15	~8%

Interpretation:

- High SLA = thin, fast-growing leaves
- High LDMC = dense, slow-decomposing tissue
- C4 grasses tend to have lower SLA, higher LDMC, thicker leaves than C3

Chemical Traits

Variable	Units	Description	Typical Range	Missing
nitrogen_content	% dry mass	Leaf nitrogen concentration	1.0-4.5	~14%
carbon_content	% dry mass	Leaf carbon concentration	42-48	~14%
C_N_ratio	dimensionless	Carbon to nitrogen ratio	10-40	~14%
chlorophyll_content	varies*	Total chlorophyll	varies	~16%
carotenoid_content	µg/cm ²	Carotenoid pigments	varies	~20%

Notes:

- Nitrogen and carbon have identical missing patterns (measured together on CN analyzer)
- Chlorophyll units unclear - could be µmol/m² or SPAD units
- Higher nitrogen = more photosynthetic capacity

Water Relations

Variable	Units	Description	Typical Range	Missing
water_content	% or proportion	Leaf water content	50-80%	~11%
EWT	g/cm ²	Equivalent Water Thickness	0.01-0.03	~17%

EWT = water mass per unit leaf area, important for SWIR spectral features

Missing Value Summary

What NA means:

- Measurement was not obtained for that sample
- Could be due to: insufficient sample material, measurement failure, or trait not measured for that species

Missing value patterns:

- Taxonomic/site data: 0% missing (complete)
- Morphological traits: 8-18% missing
- Chemical traits: 14-20% missing
- Paired traits (N and C): same samples missing

No pattern by species or site - missing data appears random

Data Quality Checks Done

During curation, I checked:

1. **Range validation:** All values fall within biologically plausible ranges
2. **Join integrity:** All 426 sample_ids match perfectly between files
3. **Outlier detection:** Flagged 3-5 potential outliers (very high SLA or chlorophyll) - could be real or errors
4. **Missing patterns:** No systematic bias - missing data is random

Files are clean and ready to use - just be aware of the ~10-20% missing trait data.

Common Questions

Q: Why do spectral column names start with "X"?

A: CSV column names can't start with numbers, so "X" prefix is added. X650.5 = 650.5 nm.

Q: What scale are reflectance values on?

A: 0-1 scale (proportion, not percentage). 0.5 = 50% reflectance.

Q: Why are some SWIR values missing?

A: Atmospheric water absorption around 1400nm and 1900nm reduces signal quality. Some values removed during QC.

Q: Are these biological or technical replicates?

A: Each row is one individual leaf from one plant (biological replicates). Not clear if multiple scans were averaged per leaf.

Q: What do I do with missing trait data?

A: Either analyze complete cases only (use `na.omit()` in R) or use imputation methods if needed.

Key Relationships to Expect

Based on leaf economics theory:

- **SLA and LDMC:** Strong negative correlation ($r \approx -0.8$)
- **SLA and nitrogen:** Positive correlation
- **Chlorophyll and nitrogen:** Positive correlation
- **NIR reflectance and leaf thickness:** Positive correlation
- **SWIR reflectance and water content:** Negative correlation

Tips for Using This Data

1. **Start with complete cases** for initial analyses to avoid missing data complications
2. **Subset by spectral region** (VIS, NIR, SWIR) for faster processing
3. **Check for C3/C4 differences** - this is a major source of variation
4. **Watch out for site effects** even though lineage is stronger

5. **Consider smoothing spectra** in noisy regions (SWIR)

References

For full methodological details, see:

Pau, S., et al. (2025). Hyperspectral leaf reflectance of grasses varies with evolutionary lineage more than with site. *Ecosphere* 16(3): e70257.